fiber bundle terminology this subspace $L^\infty \subset \Gamma X$ is a section of the bundle. Let ΛX denote the quotient $\Gamma X/L^\infty$ obtained by collapsing the section L^∞ to a point. Note that the fibers $X^{\wedge p}$ in ΓX are still embedded in the quotient ΛX since each fiber meets the section L^∞ in a single point.

If we replace S^∞ by S^1 in these definition, we get subspaces $\Gamma^1 X \subset \Gamma X$ and $\Lambda^1 X \subset \Lambda X$. All these spaces have natural CW structures if X is a CW complex having x_0 as a 0-cell. Namely, L^∞ is given its standard CW structure with one cell in each dimension. This is lifted to a CW structure on S^∞ with p cells in each dimension, and then T freely permutes the product cells of $S^\infty \times X^{\wedge p}$ so there is induced a quotient CW structure on ΓX. The section $L^\infty \subset \Gamma X$ is a subcomplex, so the quotient ΛX inherits a CW structure from ΓX. For example, if the n-skeleton of X is S^n with its usual CW structure, then the pn-skeleton of ΛX is S^{pn} with its usual CW structure.

We remark also that Γ, Γ^1, Λ, and Λ^1 are functors: A map $f:(X,x_0)\to(Y,y_0)$ induces maps $\Gamma f:\Gamma X\to\Gamma Y$, etc., in the evident way.

For brevity we write $H^*(-;\mathbb{Z}_p)$ simply as $H^*(-)$. For $n>0$ let K_n denote a CW complex $K(\mathbb{Z}_p,n)$ with $(n-1)$-skeleton a point and n-skeleton S^n. Let $\iota \in H^n(K_n)$ be the canonical fundamental class described in the discussion following Theorem 4.57. It will be notationally convenient to regard an element $\alpha \in H^n(X)$ also as a map $\alpha: X\to K_n$ such that $\alpha^*(\iota)=\alpha$. Here we are assuming X is a CW complex.

From §3.2 we have a reduced p-fold cross product $\widetilde{H}^*(X)^{\otimes p}\to\widetilde{H}^*(X^{\wedge p})$ where $\widetilde{H}^*(X)^{\otimes p}$ denotes the p-fold tensor product of $\widetilde{H}^*(X)$ with itself. This cross product map $\widetilde{H}^*(X)^{\otimes p}\to\widetilde{H}^*(X^{\wedge p})$ is an isomorphism since we are using $\mathbb{Z}_p$ coefficients. With this isomorphism in mind, we will use the notation $\alpha_1\otimes\cdots\otimes\alpha_p$ rather than $\alpha_1\times\cdots\times\alpha_p$ for p-fold cross products in $\widetilde{H}^*(X^{\wedge p})$. In particular, for each element $\alpha\in H^n(X)$, $n>0$, we have its p-fold cross product $\alpha^{\otimes p}\in\widetilde{H}^*(X^{\wedge p})$. Our first task will be to construct an element $\lambda(\alpha)\in H^{pn}(\Lambda X)$ restricting to $\alpha^{\otimes p}$ in each fiber $X^{\wedge p}\subset\Lambda X$. By naturality it will suffice to construct $\lambda(\iota)\in H^{pn}(\Lambda K_n)$.

The key point in the construction of $\lambda(\iota)$ is the fact that $T^*(\iota^{\otimes p})=\iota^{\otimes p}$. In terms of maps $K_n^{\wedge p}\to K_{pn}$, this says the composition $\iota^{\otimes p}\,T$ is homotopic to $\iota^{\otimes p}$, preserving basepoints. Such a homotopy can be constructed as follows. The pn-skeleton of $K_n^{\wedge p}$ is $(S^n)^{\wedge p}=S^{pn}$, with T permuting the factors cyclically. Thinking of S^n as $(S^1)^{\wedge n}$, the permutation T is a product of $(p-1)n$ transpositions of adjacent factors, so T has degree $(-1)^{(p-1)n}$ on S^{pn}. If p is odd, this degree is $+1$, so the restriction of T to this skeleton is homotopic to the identity, hence $\iota^{\otimes p}\,T$ is homotopic to $\iota^{\otimes p}$ on this skeleton. This conclusion also holds when $p=2$, signs being irrelevant in this case since we are dealing with maps $S^{2n}\to K_{2n}$ and $\pi_{2n}(K_{2n})=\mathbb{Z}_2$. Having a homotopy $\iota^{\otimes p}\,T\simeq\iota^{\otimes p}$ on the pn-skeleton, there are no obstructions to extending the homotopy over all higher-dimensional cells $e^i\times(0,1)$ since $\pi_i(K_{pn})=0$ for $i>pn$.

The homotopy $\iota^{\otimes p}\,T\simeq\iota^{\otimes p}:K_n^{\wedge p}\to K_{pn}$ defines a map $\Gamma^1 K_n\to K_{pn}$ since $\Gamma^1 X$ is the quotient of $I\times X^{\wedge p}$ under the identifications $(0,x)\sim(1,T(x))$. The homotopy is basepoint-preserving, so the map $\Gamma^1 K_n \to K_{pn}$ passes down to a quotient map $\lambda_1 : \Lambda^1 K_n \to K_{pn}$. Since K_n is obtained from S^n by attaching cells of dimension greater than n, ΛK_n is obtained from $\Lambda^1 K_n$ by attaching cells of dimension greater than $pn+1$. There are then no obstructions to extending λ_1 to a map $\lambda : \Lambda K_n \to K_{pn}$ since $\pi_i(K_{pn}) = 0$ for $i > pn$.

The map λ gives the desired element $\lambda(\iota) \in H^{pn}(\Lambda K_n)$ since the restriction of λ to each fiber $K_n^{\wedge p}$ is homotopic to $\iota^{\otimes p}$. Note that this property determines λ uniquely up to homotopy since the restriction map $H^{pn}(\Lambda K_n) \to H^{pn}(K_n^{\wedge p})$ is injective, the pn-skeleton of ΛK_n being contained in $K_n^{\wedge p}$. We shall have occasion to use this argument again in the proof, so we refer to it as 'the uniqueness argument.'

For any $\alpha \in H^n(X)$ let $\lambda(\alpha)$ be the composition $\Lambda X \xrightarrow{\Lambda\alpha} \Lambda K_n \xrightarrow{\lambda} K_{pn}$. This restricts to $\alpha^{\otimes p}$ in each fiber $X^{\wedge p}$ since $\Lambda\alpha$ restricts to $\alpha^{\otimes p}$ in each fiber.

Now we are ready to define some cohomology operations. There is an inclusion $L^\infty \times X \hookrightarrow \Gamma X$ as the quotient of the diagonal embedding $S^\infty \times X \hookrightarrow S^\infty \times X^{\wedge p}$, $(s,x) \mapsto (s,x,\cdots,x)$. Composing with the quotient map $\Gamma X \to \Lambda X$, we get a map $\nabla : L^\infty \times X \to \Lambda X$ inducing $\nabla^* : H^*(\Lambda X) \to H^*(L^\infty \times X) \approx H^*(L^\infty) \otimes H^*(X)$. For each $\alpha \in H^n(X)$ the element $\nabla^*(\lambda(\alpha)) \in H^{pn}(L^\infty \times X)$ may be written in the form

$$\nabla^*(\lambda(\alpha)) = \textstyle\sum_i \omega_{(p-1)n-i} \otimes \theta_i(\alpha)$$

where ω_j is a generator of $H^j(L^\infty)$ and $\theta_i(\alpha) \in H^{n+i}(X)$. Thus θ_i increases dimension by i. When $p = 2$ there is no ambiguity about ω_j. For odd p we choose ω_1 to be the class dual to the 1-cell of L^∞ in its standard cell structure, then we take ω_2 to be the Bockstein $\beta\omega_1$ and we set $\omega_{2j} = \omega_2^j$ and $\omega_{2j+1} = \omega_1\omega_2^j$.

It is clear that θ_i is a cohomology operation since $\theta_i(\alpha) = \alpha^*(\theta_i(\iota))$. For $p = 2$ we set $Sq^i(\alpha) = \theta_i(\alpha)$. For odd p we will show that $\theta_i = 0$ unless $i = 2k(p-1)$ or $2k(p-1)+1$. The operation P^k will be defined to be a certain constant times $\theta_{2k(p-1)}$, and $\theta_{2k(p-1)+1}$ will be a constant times βP^k, for β the mod p Bockstein.

Theorem 4L.12. *The operations Sq^i satisfy the properties* (1)–(7).

Proof: We have already observed that the θ_i's are cohomology operations, so property (1) holds. Because cohomology operations cannot decrease dimension, we have $\theta_i = 0$ for $i < 0$. Also, $\theta_i = 0$ for $i > (p-1)n$ since the factor $\omega_{(p-1)n-i}$ vanishes in this case. The basic property that $\lambda(\alpha)$ restricts to $\alpha^{\otimes p}$ in each fiber implies that $\theta_{(p-1)n}(\alpha) = \alpha^p$ since $\omega_0 = 1$. This gives property (5).

To show the additivity property (2) we will show $\lambda(\alpha+\beta) = \lambda(\alpha) + \lambda(\beta)$ using the following diagram:

$$\begin{tikzcd}
\Lambda X \arrow[r, "\Lambda(\Delta)"] \arrow[dr, "\Delta"'] & \Lambda(X \times X) \arrow[r, "\Lambda(\alpha\times\beta)"] \arrow[d] & \Lambda(K_n \times K_n) \arrow[r, "\Lambda(\iota\otimes 1 + 1\otimes\iota)"] \arrow[d] \arrow[dr, "\lambda(\iota\otimes 1+1\otimes\iota)"] & \Lambda K_n \arrow[d, "\lambda"] \\
& \Lambda X \times \Lambda X \arrow[r, "\Lambda\alpha\times\Lambda\beta"] & \Lambda K_n \times \Lambda K_n \arrow[r, "\lambda\otimes 1 + 1\otimes\lambda"] & K_{pn}
\end{tikzcd}$$

Here Δ is a generic symbol for diagonal maps $x \mapsto (x, x)$. These relate cross product to cup product via $\Delta^*(\varphi \otimes \psi) = \varphi \smile \psi$. The two unlabeled vertical maps are induced by $(s, x_1, y_1, \cdots, x_p, y_p) \mapsto (s, x_1, \cdots, x_p, s, y_1, \cdots, y_p)$. It is obvious that the central square in the diagram commutes, and also the left and upper right triangles. To show that the remaining triangle commutes it suffices to show this when we restrict to the fiber $(K_n \times K_n)^{\wedge p}$ by the uniqueness argument used earlier, since $(K_n \times K_n)^{\wedge p}$ contains the pn-skeleton of $\Lambda(K_n \times K_n)$. On the fiber the two routes to K_{pn} are $\iota^{\otimes p} \otimes 1 + 1 \otimes \iota^{\otimes p}$ and $(\iota \otimes 1 + 1 \otimes \iota)^{\otimes p}$. These are equal since raising to the p^{th} power is an additive homomorphism of the field $\mathbb{Z}_p$.

The composition $\Lambda X \to \Lambda K_n$ is $\Lambda(\alpha + \beta)$, so the composition $\Lambda X \to K_{pn}$ across the top of the diagram is $\lambda(\alpha + \beta)$. Going from ΛX to K_{pn} across the bottom of the diagram gives $\lambda(\alpha) + \lambda(\beta)$ since the second and third steps of this composition give $\lambda(\alpha) \otimes 1 + 1 \otimes \lambda(\beta)$ and $\Delta^*(\varphi \otimes \psi) = \varphi \smile \psi$. Thus $\lambda(\alpha + \beta) = \lambda(\alpha) + \lambda(\beta)$, which implies that each θ_i is an additive homomorphism.

Next we turn to the Cartan formula. For any prime p we will show that $\lambda(\alpha \smile \beta) = (-1)^{p(p-1)mn/2} \lambda(\alpha)\lambda(\beta)$ for $m = |\alpha|$ and $n = |\beta|$. This implies (3) when $p = 2$ since if we let $\omega = \omega_1$, hence $\omega_j = \omega^j$, then

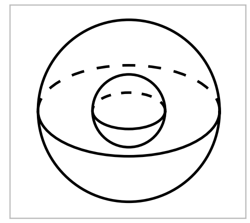
$$\begin{aligned}
\textstyle\sum_i Sq^i(\alpha \smile \beta) \otimes \omega^{n+m-i} &= \nabla^*\big(\lambda(\alpha \smile \beta)\big) = \nabla^*\big(\lambda(\alpha) \smile \lambda(\beta)\big) \\
&= \nabla^*\big(\lambda(\alpha)\big) \smile \nabla^*\big(\lambda(\beta)\big) \\
&= \textstyle\sum_j Sq^j(\alpha) \otimes \omega^{n-j} \smile \sum_k Sq^k(\beta) \otimes \omega^{m-k} \\
&= \textstyle\sum_i \Big(\sum_{j+k=i} Sq^j(\alpha) \smile Sq^k(\beta)\Big) \otimes \omega^{n+m-i}
\end{aligned}$$

To obtain the formula $\lambda(\alpha \smile \beta) = \pm\lambda(\alpha) \smile \lambda(\beta)$ we use a diagram rather like the previous one:

$$\begin{tikzcd}
\Lambda X \arrow[r, "\Lambda(\Delta)"] \arrow[dr, "\Delta"'] & \Lambda(X \wedge X) \arrow[r, "\Lambda(\alpha \wedge \beta)"] \arrow[d] & \Lambda(K_m \wedge K_n) \arrow[r, "\Lambda(\iota_m \otimes \iota_n)"] \arrow[d] \arrow[dr, "\lambda(\iota_m \otimes \iota_n)"] & \Lambda K_{m+n} \arrow[d, "\lambda"] \\
& \Lambda X \wedge \Lambda X \arrow[r, "\Lambda\alpha \wedge \Lambda\beta"] & \Lambda K_m \wedge \Lambda K_n \arrow[r, "\lambda(\iota_m) \otimes \lambda(\iota_n)"] & K_{pm+pn}
\end{tikzcd}$$

The composition $\Lambda X \to K_{pm+pn}$ going across the top of the diagram is $\lambda(\alpha \smile \beta)$ since the map $\Lambda X \to \Lambda K_{m+n}$ is $\Lambda(\alpha \smile \beta)$. The composition $\Lambda X \wedge \Lambda X \to K_{pm+pn}$ is $\lambda(\alpha) \otimes \lambda(\beta)$ so the composition $\Lambda X \to K_{pm+pn}$ across the bottom of the diagram is $\lambda(\alpha) \smile \lambda(\beta)$. Commutativity of the square and the upper two triangles is obvious from the definitions. It remains to see that the lower triangle commutes up to the sign $(-1)^{p(p-1)mn/2}$. Since $(K_m \wedge K_n)^{\wedge p}$ includes the $(pm + pn)$-skeleton of $\Lambda(K_m \wedge K_n)$, restriction to this fiber is injective on H^{pm+pn}. On this fiber the two routes around the triangle give $(\iota_m \otimes \iota_n)^{\otimes p}$ and $\iota_m^{\otimes p} \otimes \iota_n^{\otimes p}$. These differ by a permutation that is the product of $(p-1) + (p-2) + \cdots + 1 = p(p-1)/2$ transpositions of adjacent factors. Since ι_m and ι_n have dimensions m and n, this permutation introduces a sign $(-1)^{p(p-1)mn/2}$ by the commutativity property of cup product. This finishes the verification of the Cartan formula when $p = 2$.

Before proceeding further we need to make an explicit calculation to show that Sq^0 is the identity on $H^1(S^1)$. Viewing S^1 as the one-point compactification of $\mathbb{R}$, with the point at infinity as the basepoint, the 2-sphere $S^1 \wedge S^1$ becomes the one-point compactification of $\mathbb{R}^2$. The map $T: S^1 \wedge S^1 \to S^1 \wedge S^1$ then corresponds to reflecting $\mathbb{R}^2$ across the line $x = y$, so after a rotation of coordinates this becomes reflection of S^2 across the equator. Hence $\Gamma^1 S^1$ is obtained from the shell $I \times S^2$ by identifying its inner and outer boundary spheres via a reflection across the equator. The diagonal $\mathbb{R}P^1 \times S^1 \subset \Gamma^1 S^1$ is a torus, obtained from the equatorial annulus $I \times S^1 \subset I \times S^2$ by identifying the two ends via the identity map since the equator is fixed by the reflection. This $\mathbb{R}P^1 \times S^1$ represents the same element of $H_2(\Gamma^1 S^1; \mathbb{Z}_2)$ as the fiber sphere $S^1 \wedge S^1$ since the upper half of the shell is a 3-cell whose mod 2 boundary in $\Gamma^1 S^1$ is the union of these two surfaces.



For a generator $\alpha \in H^1(S^1)$, consider the element $\nabla^*(\lambda(\alpha))$ in $H^2(\mathbb{R}P^\infty \times S^1) \approx \mathrm{Hom}(H_2(\mathbb{R}P^\infty \times S^1; \mathbb{Z}_2), \mathbb{Z}_2)$. A basis for $H_2(\mathbb{R}P^\infty \times S^1; \mathbb{Z}_2)$ is represented by $\mathbb{R}P^2 \times \{x_0\}$ and $\mathbb{R}P^1 \times S^1$. A cocycle representing $\nabla^*(\lambda(\alpha))$ takes the value 0 on $\mathbb{R}P^2 \times \{x_0\}$ since $\mathbb{R}P^\infty \times \{x_0\}$ collapses to a point in ΛS^1 and $\lambda(\alpha)$ lies in $H^2(\Lambda S^1)$. On $\mathbb{R}P^1 \times S^1$, $\nabla^*(\lambda(\alpha))$ takes the value 1 since when $\lambda(\alpha)$ is pulled back to ΓS^1 it takes the same value on the homologous cycles $\mathbb{R}P^1 \times S^1$ and $S^1 \wedge S^1$, namely 1 by the defining property of $\lambda(\alpha)$ since $\alpha \otimes \alpha \in H^2(S^1 \wedge S^1)$ is a generator. Thus $\nabla^*(\lambda(\alpha)) = \omega_1 \otimes \alpha$ and hence $Sq^0(\alpha) = \alpha$ by the definition of Sq^0.

We use this calculation to prove that Sq^i commutes with the suspension σ, where σ is defined by $\sigma(\alpha) = \varepsilon \otimes \alpha \in H^*(S^1 \wedge X)$ for ε a generator of $H^1(S^1)$ and $\alpha \in H^*(X)$. We have just seen that $Sq^0(\varepsilon) = \varepsilon$. By (5), $Sq^1(\varepsilon) = \varepsilon^2 = 0$ and $Sq^i(\varepsilon) = 0$ for $i > 1$. The Cartan formula then gives $Sq^i(\sigma(\alpha)) = Sq^i(\varepsilon \otimes \alpha) = \sum_j Sq^j(\varepsilon) \otimes Sq^{i-j}(\alpha) = \varepsilon \otimes Sq^i(\alpha) = \sigma(Sq^i(\alpha))$.

From this it follows that Sq^0 is the identity on $H^n(S^n)$ for all $n > 0$. Since S^n is the n-skeleton of K_n, this implies that Sq^0 is the identity on the fundamental class ι_n, hence Sq^0 is the identity on all positive-dimensional classes.

Property (7) is proved similarly: Sq^1 coincides with the Bockstein β on the generator $\omega \in H^1(\mathbb{R}P^2)$ since both equal ω^2. Hence $Sq^1 = \beta$ on the iterated suspensions of ω, and the n-fold suspension of $\mathbb{R}P^2$ is the $(n+2)$-skeleton of K_{n+1}. □

Theorem 4L.13. *The Adem relations hold for Steenrod squares.*

Proof: The idea is to imitate the construction of ΛX using $\mathbb{Z}_p \times \mathbb{Z}_p$ in place of $\mathbb{Z}_p$. The Adem relations will then arise from the symmetry of $\mathbb{Z}_p \times \mathbb{Z}_p$ interchanging the two factors.

The group $\mathbb{Z}_p\times\mathbb{Z}_p$ acts on $S^\infty\times S^\infty$ via $(g,h)(s,t)=(g(s),h(t))$, with quotient $L^\infty\times L^\infty$. There is also an action of $\mathbb{Z}_p\times\mathbb{Z}_p$ on $X^{\wedge p^2}$, obtained by writing points of $X^{\wedge p^2}$ as p^2-tuples (x_{ij}) with subscripts i and j varying from 1 to p, and then letting the first $\mathbb{Z}_p$ act on the first subscript and the second $\mathbb{Z}_p$ act on the second. Factoring out the diagonal action of $\mathbb{Z}_p\times\mathbb{Z}_p$ on $S^\infty\times S^\infty\times X^{\wedge p^2}$ gives a quotient space Γ_2X. This projects to $L^\infty\times L^\infty$ with a section, and collapsing the section gives Λ_2X. The fibers of the projection $\Lambda_2X\to L^\infty\times L^\infty$ are $X^{\wedge p^2}$ since the action of $\mathbb{Z}_p\times\mathbb{Z}_p$ on $S^\infty\times S^\infty$ is free. We could also obtain Λ_2X from the product $S^\infty\times S^\infty\times X^{p^2}$ by first collapsing the subspace of points having at least one X coordinate equal to the basepoint x_0 and then factoring out the $\mathbb{Z}_p\times\mathbb{Z}_p$ action.

It will be useful to compare Λ_2X with $\Lambda(\Lambda X)$. The latter space is the quotient of $S^\infty\times(S^\infty\times X^p)^p$ in which one first identifies all points having at least one X coordinate equal to x_0 and then one factors out by an action of the wreath product $\mathbb{Z}_p\wr\mathbb{Z}_p$, the group of order p^{p+1} defined by a split exact sequence $0\to\mathbb{Z}_p^p\to\mathbb{Z}_p\wr\mathbb{Z}_p\to\mathbb{Z}_p\to 0$ with conjugation by the quotient group $\mathbb{Z}_p$ given by cyclic permutations of the p $\mathbb{Z}_p$ factors of $\mathbb{Z}_p^p$. In the coordinates $(s,t_1,x_{11},\cdots,x_{1p},\cdots,t_p,x_{p1},\cdots,x_{pp})$ the i^{th} factor $\mathbb{Z}_p$ of $\mathbb{Z}_p^p$ acts in the block $(t_i,x_{i1},\cdots,x_{ip})$, and the quotient $\mathbb{Z}_p$ acts by cyclic permutation of the index i and by rotation in the s coordinate. There is a natural map $\Lambda_2X\to\Lambda(\Lambda X)$ induced by $(s,t,x_{11},\cdots,x_{1p},\cdots,x_{p1},\cdots,x_{pp})\mapsto(s,t,x_{11},\cdots,x_{1p},\cdots,t,x_{p1},\cdots,x_{pp})$. In Λ_2X one is factoring out by the action of $\mathbb{Z}_p\times\mathbb{Z}_p$. This is the subgroup of $\mathbb{Z}_p\wr\mathbb{Z}_p$ obtained by restricting the action of the quotient $\mathbb{Z}_p$ on $\mathbb{Z}_p^p$ to the diagonal subgroup $\mathbb{Z}_p\subset\mathbb{Z}_p^p$, where this action becomes trivial so that one has the direct product $\mathbb{Z}_p\times\mathbb{Z}_p$.

Since it suffices to prove that the Adem relations hold on the class $\iota\in H^n(K_n)$, we take $X=K_n$. There is a map $\lambda_2:\Lambda_2K_n\to K_{p^2n}$ restricting to $\iota^{\otimes p^2}$ in each fiber. This is constructed by the same method used to construct λ. One starts with a map representing $\iota^{\otimes p^2}$ in a fiber, then extends this over the part of Λ_2K_n projecting to the 1-skeleton of $L^\infty\times L^\infty$, and finally one extends inductively over higher-dimensional cells of Λ_2K_n using the fact that K_{p^2n} is an Eilenberg–MacLane space. The map λ_2 fits into the diagram at the right, where ∇_2 is induced by the map $(s,t,x)\mapsto(s,t,x,\cdots,x)$ and the unlabeled map is the one defined above. It is clear that the square commutes.

$$\begin{tikzcd}
L^\infty\times L^\infty\times K_n \arrow[r, "\nabla_2"] \arrow[d, "\mathbb{1}\times\nabla"] & \Lambda_2K_n \arrow[r, "\lambda_2"] \arrow[d] & K_{p^2n} \\
L^\infty\times\Lambda K_n \arrow[r, "\nabla"] & \Lambda(\Lambda K_n) \arrow[ur, "\lambda(\lambda)"'] &
\end{tikzcd}$$

Commutativity of the triangle up to homotopy follows from the fact that λ_2 is uniquely determined, up to homotopy, by its restrictions to fibers.

The element $\nabla_2^*\lambda_2^*(\iota)$ may be written in the form $\sum_{r,s}\omega_r\otimes\omega_s\otimes\varphi_{rs}$, and we claim that the elements φ_{rs} satisfy the symmetry relation $\varphi_{rs}=(-1)^{rs+p(p-1)n/2}\varphi_{sr}$. To verify this we use the commutative diagram at the right where the map τ on the left switches the two L^∞ factors and the τ on

$$\begin{tikzcd}
L^\infty\times L^\infty\times K_n \arrow[r, "\nabla_2"] \arrow[d, "\tau"] & \Lambda_2K_n \arrow[d, "\tau"] \\
L^\infty\times L^\infty\times K_n \arrow[r, "\nabla_2"] & \Lambda_2K_n
\end{tikzcd}$$

the right is induced by switching the two S^∞ factors of $S^\infty\times S^\infty\times K_n^{\wedge p^2}$ and permuting the K_n factors of the smash product by interchanging the two subscripts in p^2-tuples (x_{ij}). This permutation is a product of $p(p-1)/2$ transpositions, one for each pair (i,j) with $1\le i<j\le p$, so in a fiber the class $\iota^{\otimes p^2}$ is sent to $(-1)^{p(p-1)n/2}\iota^{\otimes p^2}$. By the uniqueness property of λ_2 this means that $\tau^*\lambda_2^*(\iota)=(-1)^{p(p-1)n/2}\lambda_2^*(\iota)$. Commutativity of the square then gives

$$(-1)^{p(p-1)n/2}\nabla_2^*\lambda_2^*(\iota)=\nabla_2^*\tau^*\lambda_2^*(\iota)=\tau^*\nabla_2^*\lambda_2^*(\iota)=\textstyle\sum_{r,s}(-1)^{rs}\omega_s\otimes\omega_r\otimes\varphi_{rs}$$

where the last equality follows from the commutativity property of cross products. The symmetry relation $\varphi_{rs}=(-1)^{rs+p(p-1)n/2}\varphi_{sr}$ follows by interchanging the indices r and s in the last summation.

If we compute $\nabla_2^*\lambda_2^*(\iota)$ using the lower route across the earlier diagram containing the map λ_2, we obtain

$$\begin{aligned}\nabla_2^*\lambda_2^*(\iota)&=\textstyle\sum_i\omega_{(p-1)pn-i}\otimes\theta_i\Big(\sum_j\omega_{(p-1)n-j}\otimes\theta_j(\iota)\Big)\\&=\textstyle\sum_{i,j}\omega_{(p-1)pn-i}\otimes\theta_i\big(\omega_{(p-1)n-j}\otimes\theta_j(\iota)\big)\end{aligned}$$

Now we specialize to $p=2$, so $\theta_i=Sq^i$ for all i. The Cartan formula converts the last summation above into $\sum_{i,j,k}\omega^{2n-i}\otimes Sq^k(\omega^{n-j})\otimes Sq^{i-k}Sq^j(\iota)$. Plugging in the value for $Sq^k(\omega^{n-j})$ computed in the discussion preceding Example 4L.3, we obtain $\sum_{i,j,k}\binom{n-j}{k}\omega^{2n-i}\otimes\omega^{n-j+k}\otimes Sq^{i-k}Sq^j(\iota)$. To write this summation more symmetrically with respect to the two ω terms, let $n-j+k=2n-\ell$. Then we get

$$\textstyle\sum_{i,j,\ell}\binom{n-j}{n+j-\ell}\omega^{2n-i}\otimes\omega^{2n-\ell}\otimes Sq^{i+\ell-n-j}Sq^j(\iota)$$

In view of the symmetry property of φ_{rs}, which becomes $\varphi_{rs}=\varphi_{sr}$ for $p=2$, switching i and ℓ in this formula leaves it unchanged. Hence we get the relation

$$(*)\qquad \textstyle\sum_j\binom{n-j}{n+j-\ell}Sq^{i+\ell-n-j}Sq^j(\iota)=\sum_j\binom{n-j}{n+j-i}Sq^{i+\ell-n-j}Sq^j(\iota)$$

This holds for all n, i, and ℓ, and the idea is to choose these numbers so that the left side of this equation has only one nonzero term. Given integers r and s, let $n=2^r-1+s$ and $\ell=n+s$, so that $\binom{n-j}{n+j-\ell}=\binom{2^r-1-(j-s)}{j-s}$. If r is sufficiently large, this will be 0 unless $j=s$. This is because the dyadic expansion of 2^r-1 consists entirely of 1's, so the expansion of $2^r-1-(j-s)$ will have 0's in the positions where the expansion of $j-s$ has 1's, hence these positions contribute factors of $\binom{0}{1}=0$ to $\binom{2^r-1-(j-s)}{j-s}$. Thus with n and ℓ chosen as above, the relation $(*)$ becomes

$$Sq^iSq^s(\iota)=\textstyle\sum_j\binom{2^r-1+s-j}{2^r-1+s+j-i}Sq^{i+s-j}Sq^j(\iota)=\sum_j\binom{2^r+s-j-1}{i-2j}Sq^{i+s-j}Sq^j(\iota)$$

where the latter equality comes from the general relation $\binom{x}{y}=\binom{x}{x-y}$.

The final step is to show that $\binom{2^r+s-j-1}{i-2j}=\binom{s-j-1}{i-2j}$ if $i<2s$. Both of these binomial coefficients are zero if $i<2j$. If $i\ge 2j$ then we have $2j\le i<2s$, so $j<s$,

hence $s-j-1 \ge 0$. The term 2^r then makes no difference in $\binom{2^r+s-j-1}{i-2j}$ if r is large since this 2^r contributes only a single 1 to the dyadic expansion of $2^r+s-j-1$, far to the left of all the nonzero entries in the dyadic expansions of $s-j-1$ and $i-2j$.

This gives the Adem relations for the classes ι of dimension $n = 2^r-1+s$ with r large. This implies the relations hold for all classes of these dimensions, by naturality. Since we can suspend repeatedly to make any class have dimension of this form, the Adem relations must hold for all cohomology classes. $\square$

Steenrod Powers

Our remaining task is to verify the axioms and Adem relations for the Steenrod powers for an odd prime p. Unfortunately this is quite a bit more complicated than the $p = 2$ case, largely because one has to be very careful in computing the many coefficients in $\mathbb{Z}_p$ that arise. Even for the innocent-looking axiom $P^0 = \mathbb{1}$ it will take three pages to calculate the normalization constants needed to make the axiom hold. One could wish that the whole process was a lot cleaner.

Lemma 4L.14. *$\theta_i = 0$ unless $i = 2k(p-1)$ or $2k(p-1)+1$.*

Proof: The group of automorphisms of $\mathbb{Z}_p$ is the multiplicative group $\mathbb{Z}_p^*$ of nonzero elements of $\mathbb{Z}_p$. Since p is prime, $\mathbb{Z}_p$ is a field and $\mathbb{Z}_p^*$ is cyclic of order $p-1$. Let r be a generator of $\mathbb{Z}_p^*$. Define a map $\varphi : S^\infty \times X^{\wedge p} \to S^\infty \times X^{\wedge p}$ by $\varphi(s,(x_i)) = (s^r,(x_{ri}))$ where subscripts are taken mod p and s^r means raise each coordinate of s, regarded as a unit vector in $\mathbb{C}^\infty$, to the r^{th} power and renormalize the resulting vector to have unit length. Then if γ is a generator of the $\mathbb{Z}_p$ action on $S^\infty \times X^{\wedge p}$, we have $\varphi(\gamma(s,(x_i))) = \varphi(e^{2\pi/p}s,(x_{i+1})) = (e^{2r\pi/p}s^r,(x_{ri+r})) = \gamma^r(\varphi(s,(x_i)))$. This says that φ takes orbits to orbits, so φ induces maps $\varphi : \Gamma X \to \Gamma X$ and $\varphi : \Lambda X \to \Lambda X$. Restricting to the first coordinate, there is also an induced map $\varphi : L^\infty \to L^\infty$. Taking $X = K_n$, these maps fit into the diagram at the right. The square obviously commutes, and the triangle commutes up to homotopy since it suffices to verify this in the fiber $X^{\wedge p}$, and here the map φ is an even permutation of the coordinates since it has even order $p-1$, so φ preserves $\iota^{\otimes p}$.

$$\begin{tikzcd}
L^\infty \times K_n \arrow[r, "\nabla"] \arrow[d, "\varphi\times\mathbb{1}"'] & \Lambda K_n \arrow[r, "\lambda"] \arrow[d, "\varphi"'] & K_{pn} \\
L^\infty \times K_n \arrow[r, "\nabla"] & \Lambda K_n \arrow[ur, "\lambda"'] &
\end{tikzcd}$$

Commutativity of the diagram means that $\sum_i \omega_{(p-1)n-i} \otimes \theta_i(\iota)$ is invariant under $\varphi^* \otimes \mathbb{1}$. The map φ induces multiplication by r in $\pi_1(L^\infty)$, hence also in $H_1(L^\infty)$ and $H^1(L^\infty;\mathbb{Z}_p)$, sending ω_1 to $r\omega_1$. Since ω_2 was chosen to be the Bockstein of ω_1, it is also multiplied by r. We chose r to have order $p-1$ in $\mathbb{Z}_p^*$, so $\varphi^*(\omega_\ell) = \omega_\ell$ only when the total number of ω_1 and ω_2 factors in ω_ℓ is a multiple of $p-1$, that is, when ℓ has the form $2k(p-1)$ or $2k(p-1)-1$. Thus in the formula $\sum_i \omega_{(p-1)n-i} \otimes \theta_i(\iota)$ we must have $\theta_i(\iota) = 0$ unless i is congruent to 0 or 1 mod $2(p-1)$. $\square$